VineWeave Canopy

A tree-like support system for climbing plants and pedestrian shade

MEMBER	RESPONSIBILITY & KEY WORK	DELIVERABLES
臧唯羽	Design · Form and parametric modelling Rhino/Grasshopper; geometry and drawings	3D model; plans, elevations and parameter diagrams
林新淇	Design · Materials and construction Material/plant selection; assembly and cost	Material rationale; build-up and cost estimate
虞寒哲	Design · Shade and application Canopy analysis; comfort mechanisms and use	Shade analysis; application and design statement
徐世程	PPT · Production and integration Structure, charts, layout and visual consistency	Integrated presentation within 20 slides
陈淼	Presentation · Script and delivery English presentation, timing and Q&A	Presentation script and live delivery

Team responsibilities follow the supplied group allocation.

A shaded pause point in public space

Combine climbing vegetation, a tree-like support and clear pedestrian circulation.

01

SOLAR EXPOSURE

Offer local shelter for short pedestrian stops.

02

CLIMBING SUPPORT

Guide vegetation across a broad, elevated crown.

03

OPEN CIRCULATION

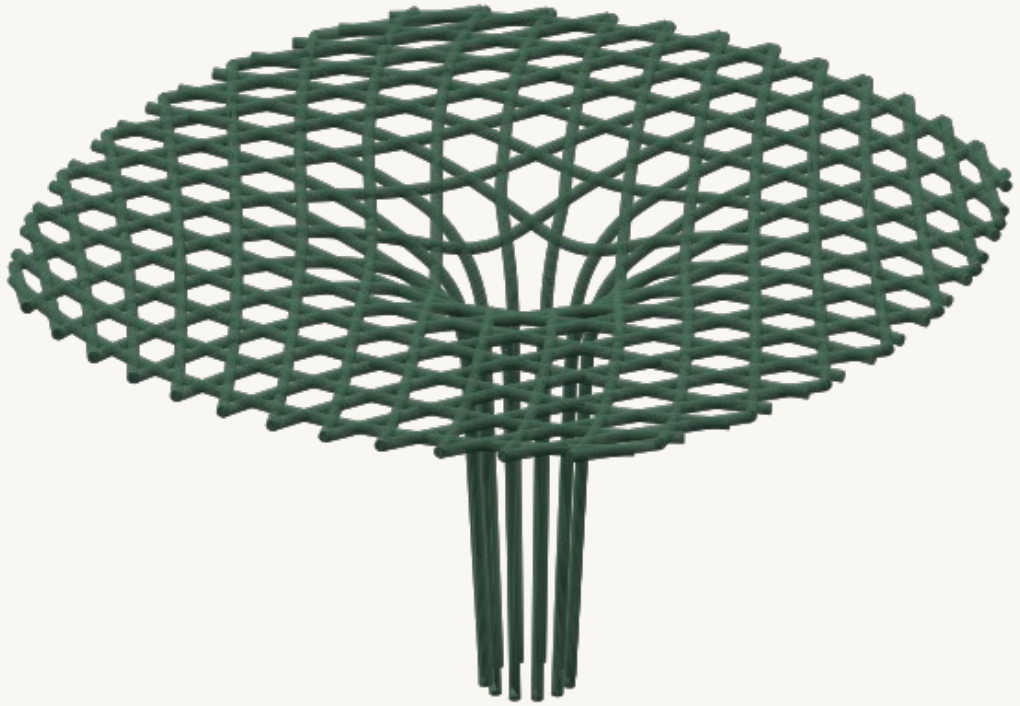
Keep the surrounding walking route accessible.

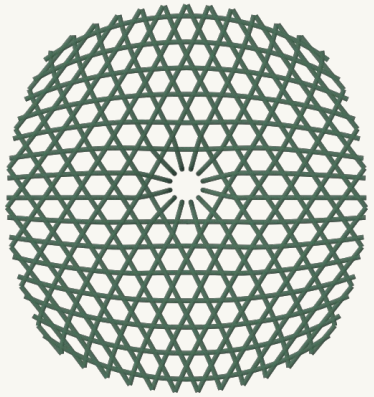
Design goal: reduce direct solar exposure while supporting everyday use.

Mechanism references: [1,2]. Project design: P4. Thermal improvement is an objective, not a measured result.

A broad crown on a compact tree-like support

Project geometry—not a certified structural or fabrication model.





PLAN



ELEVATION

6.22 × 6.66 m

Overall plan extent

59 MEMBERS

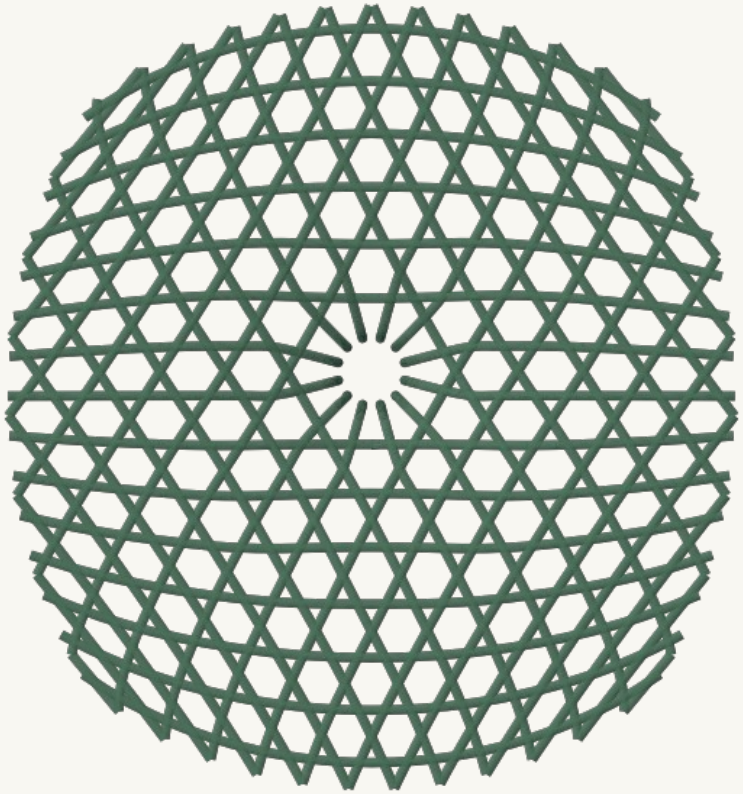
3.48 m MAX. HEIGHT

80 mm EXTERIOR DIAMETER

Source: project model and geometry records (P1). Dimensions rounded; exterior geometry diameter 80 mm.

A climbing envelope with open edges

Organise the crown, central opening and pedestrian edge as one system.



01 INTERWOVEN CROWN

Provides the proposed climbing envelope.

02 CENTRAL OPENING

Leaf-hole size is an explicit analysis assumption.

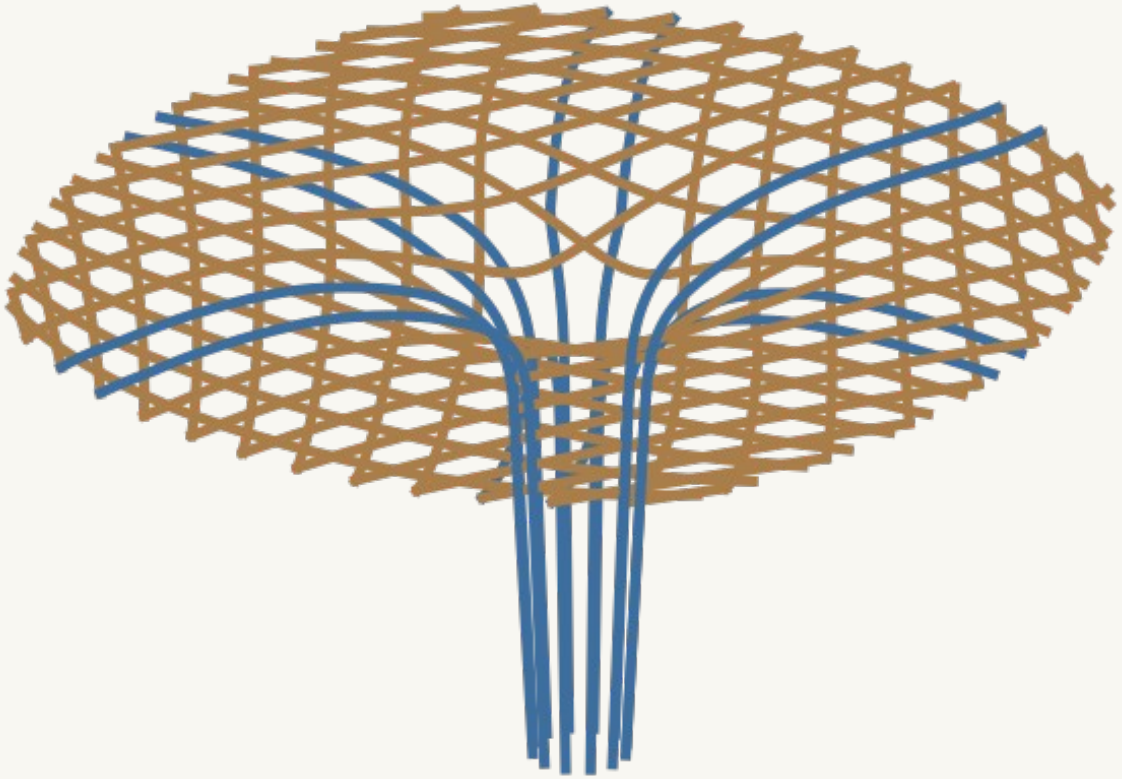
03 OPEN SIDES

Retain space for movement and potential airflow.

Sources: project geometry and proposed layout (P1, P4). The leaf envelope is hypothetical.

Assign materials by structural role

A proposed steel support zone and engineered-bamboo crown.



12 STEEL SUPPORTS

69.38 m · proposed galvanised Q355B tube

47 BAMBOO CROWN MEMBERS

219.28 m · proposed engineered bamboo

Sources: project centreline classification and material schedule (P1, P3). Proposed material assignment—not a safety certification.

Material selection balances several requirements

Preliminary screening inputs; match all bamboo properties to a confirmed product.

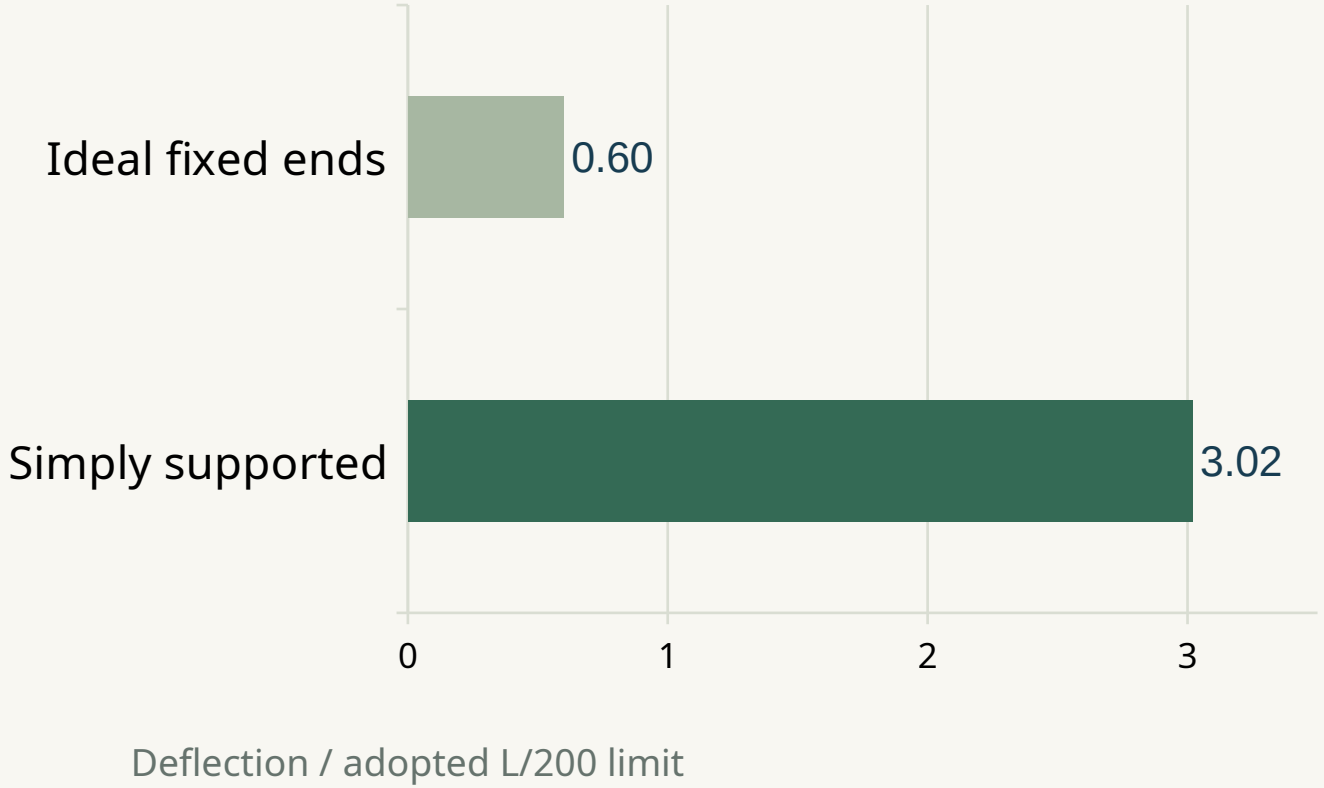
MATERIAL	λ* W/(m·K)	CARBON FACTOR* $kgCO_2e/kg$	DESIGN IMPLICATION
Galvanised steel	50	2.15	Proposed supports; protect against corrosion.
Engineered bamboo*	0.22	0.40	Proposed crown; stiffness and treatment need confirmation.
Treated timber	0.14	1.10	Screened alternative; capacity and outdoor care matter.

Product identity, strength, stiffness and carbon boundaries still require evidence.

Source: material-input workbook (P3). *Reference inputs—not independently verified product data or quotations.

Connection restraint governs deflection

A simplified beam check—not a verified response of the complete canopy.



SAME 6.65 m SPAN

Self-weight + assumed wet foliage

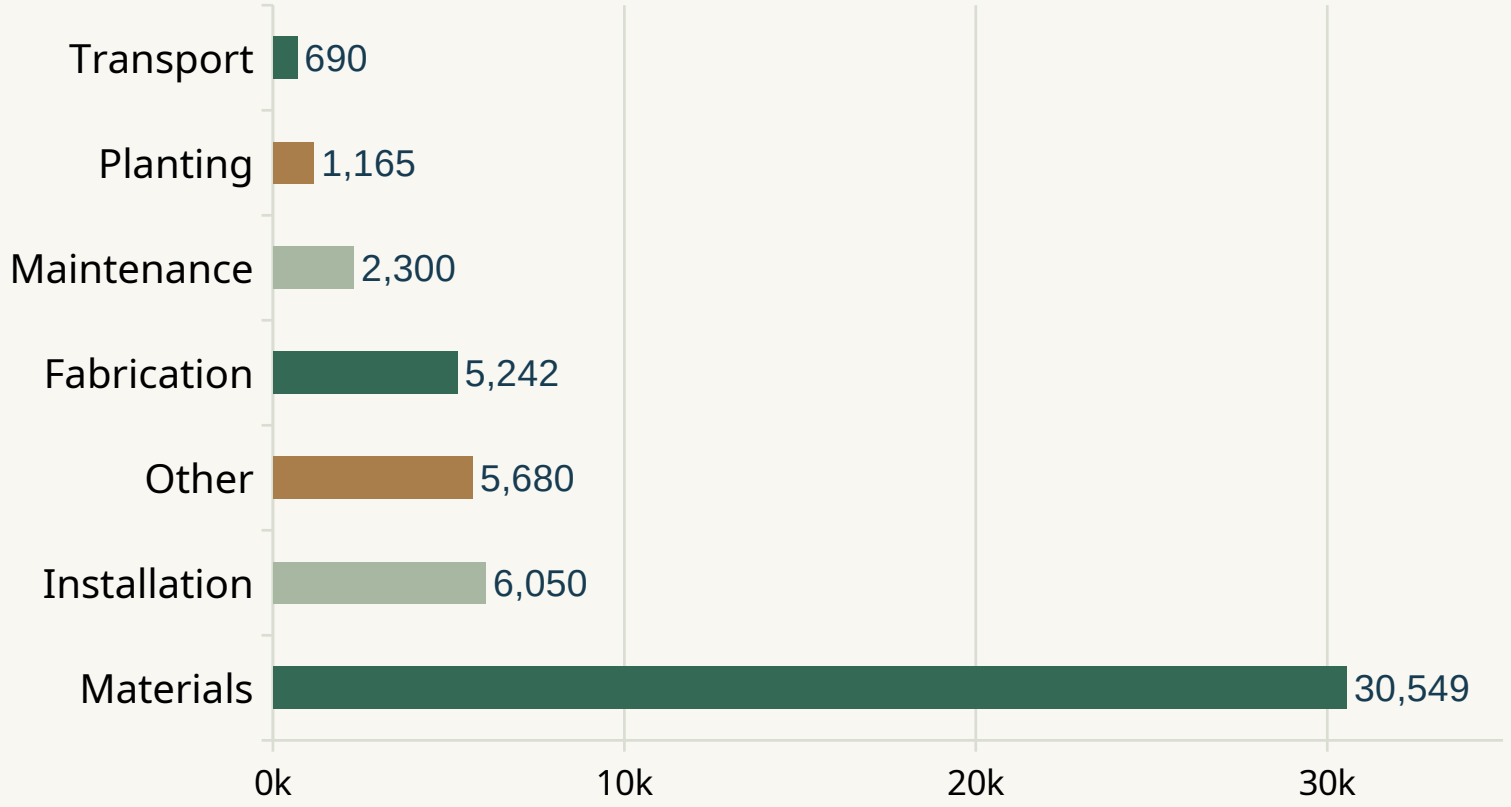
RATIO ≤ 1.0 IS THE TARGET

Actual rotational restraint is not verified.

Source: simplified deflection workbook (P3). Bamboo inputs and actual joint restraint require verification.

A preliminary budget of CNY 51,676 per unit

The total includes more than the visible structural members.



CNY 51,676

Preliminary total per unit

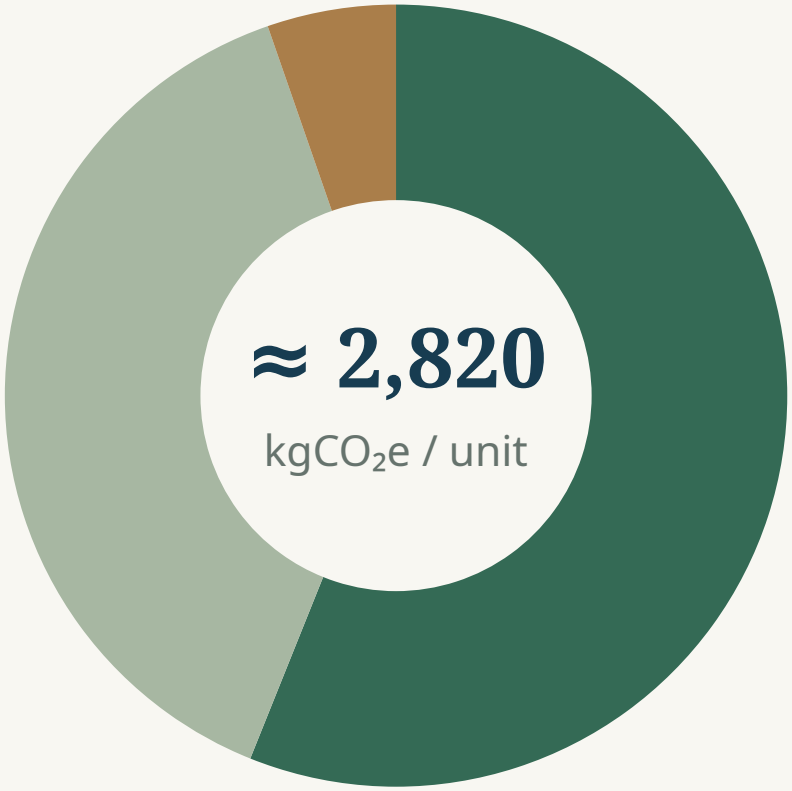
CNY 12,000

Rigid-node allowance within materials

Source: Cost-Analysis.xlsx (P3). Preliminary estimate—not supplier quotations; node price is particularly uncertain.

Track embodied carbon—and long-term care

Current factors indicate where further checking and reduction efforts should focus.



Primary members	1,581
Counterweight	1,089
Other components	150

Protect steel • Treat bamboo • Inspect joints

Source: workbook carbon estimate (P3). Preliminary factors; boundaries, processing and node inclusion need verification.

Plant choice is part of the performance design

Proposed candidates—not a confirmed site-specific planting specification.

Boston ivy

Primary candidate

Tendrils with adhesive pads

Check anchorage and seasonal cover.

Trumpet creeper

Alternative

Aerial roots

Manage vigorous growth and root space.

Rangoon creeper

Companion candidate

Twining shoots

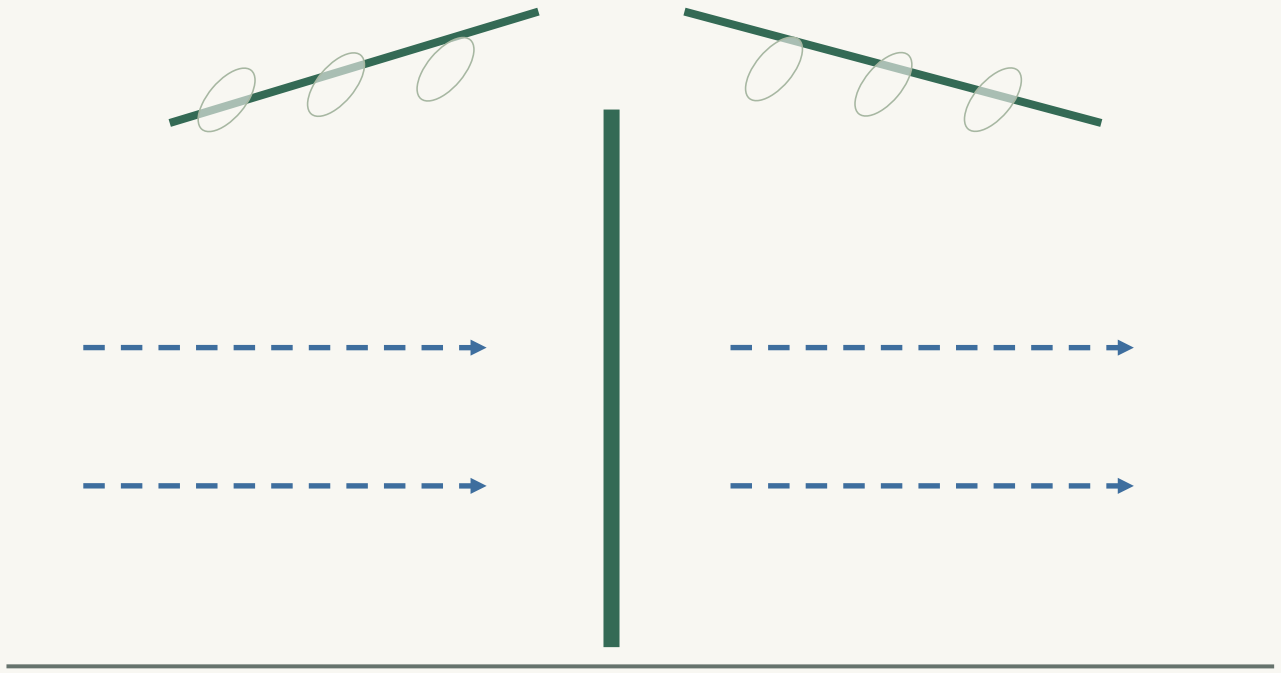
Confirm local winter survival and leaf habit.

Root space • Water supply • Training • Pruning

Source: project planting screen (P3). Local suitability, wet loads and phenology require confirmation.

Open sides and a porous crown

Shade and potential airflow paths are design intentions with different evidence needs.



CONCEPTUAL AIRFLOW

01 SHADE

Reduce direct solar exposure. [1,2]

02 OPEN EDGES

Retain potential lateral airflow paths.

03 FOLIAGE DENSITY

More cover may also increase flow resistance.

Mechanisms: [1,2]. Airflow sketch: project concept (P4), not a wind-field simulation.

Make the assumptions—and the metrics—explicit

One fixed reference zone; two different shade metrics.

30°N

Illustrative location

09:00–15:00

True solar time

23.23 m²

5 × 5 m minus r = 0.75 m

ASSUMED FOLIAGE

No site obstructions

IDEAL GEOMETRIC OVERLAP

Continuous-leaf shadow intersecting the reference zone.

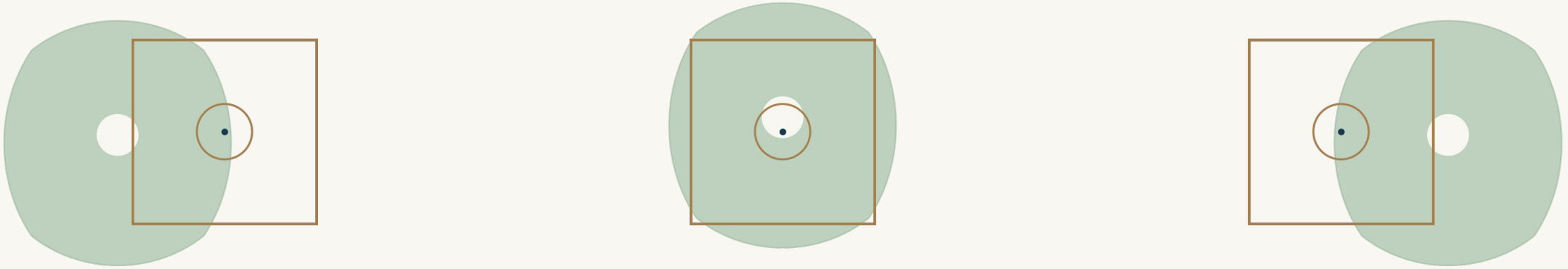
EQUIVALENT DIRECT-BEAM BLOCKAGE

Frame + leaf patches; overlap deducted; τ = 0.15.

Solar method: NOAA [3]. Reference zone and optical scenarios: project analysis (P2, P5).

Noon overlap does not mean all-day shade

Ideal continuous-leaf overlap within the fixed reference zone.



09:00 SOLAR TIME

47.2%

12:00 SOLAR TIME

98.7%

15:00 SOLAR TIME

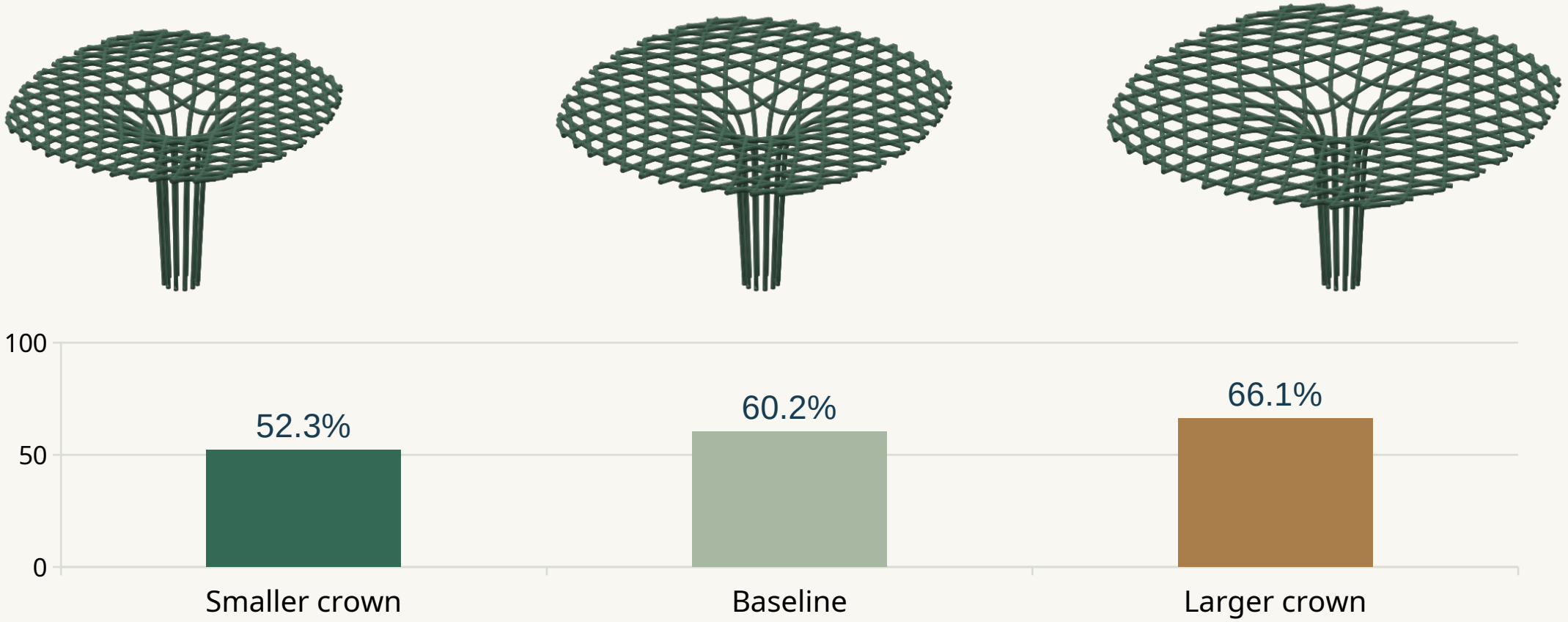
47.2%

09:00–15:00 TIME MEAN: 75.3%

Source: project geometric projection (P2). Summer solstice; 30°N; assumed leaf envelope and orientation.

Crown width changes useful shade

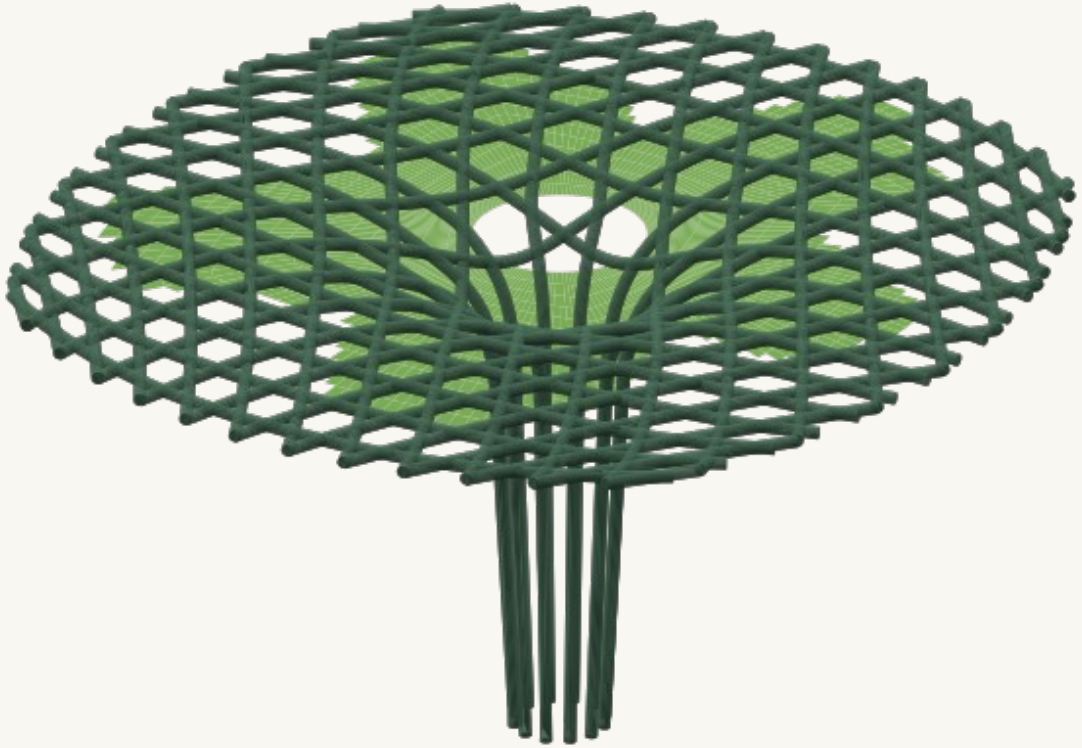
Same target and 50% plan cover; wider crowns also require more material and planted area.



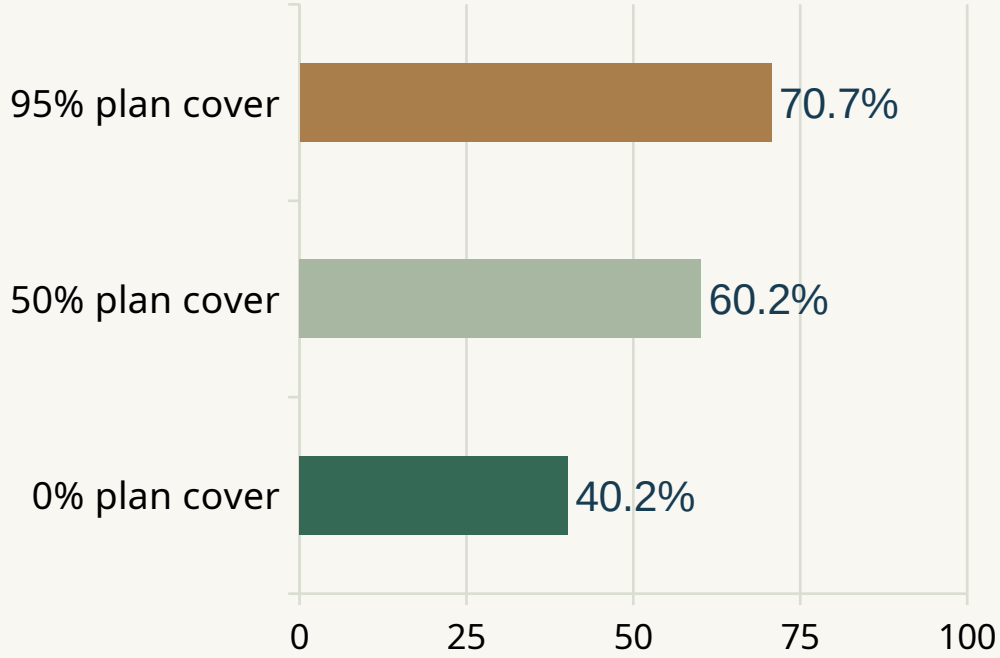
Source: parameter and spatial-cover simulation (P5). Solstice 09–15 mean; τ = 0.15; assumed conditions.

Foliage development changes blockage

Fixed structure; cover scenarios have no assigned growth months or years.



STATIC VIEW: 50% PLAN COVER • ANIMATION IN PPT

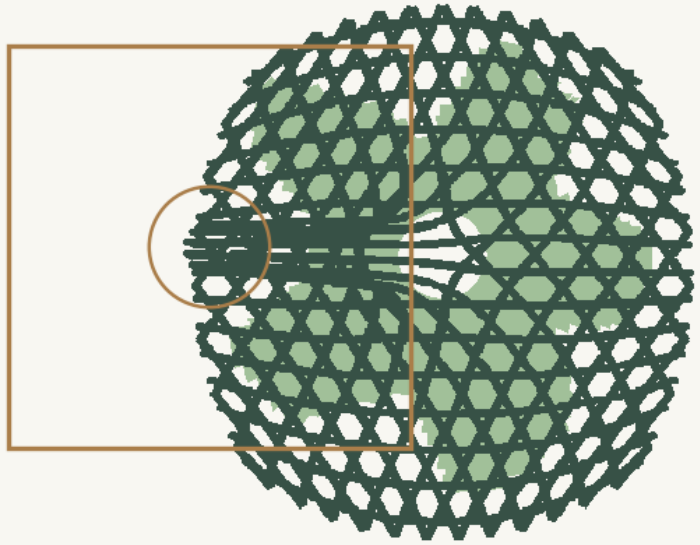


MEAN DIRECT-BEAM BLOCKAGE

Source: spatial-cover simulation (P5). Solstice 09–15 mean; frame/leaf overlap deducted; τ = 0.15.

Place seating for the occupied hours

Illustrative shade movement—not a real site or a compliance drawing.



STATIC VIEW: 15:00 • FIXED 50% PLAN COVER

01 FOLLOW THE SHADE

Use occupied-hour footprints, not only noon.

02 KEEP ROUTES CLEAR

Separate stopping places from through movement.

03 ALLOW FOR PLANT CARE

Reserve roots, water and maintenance access.

Sources: project design and fixed-50%-cover projection (P4, P5). Proposed comfort validation may follow [4].

Prefabricate components; manage site complexity

Proposed assembly logic—not a tested construction programme.

01

PREFABRICATE

Form, treat and number

›

02

POSITION BASE

Lift and align

›

03

ASSEMBLE

Support and crown

›

04

GROUT & CURE

Verify node details

›

05

INSPECT & PLANT

Check, train and irrigate

14 PERSON-DAYS

Current labour estimate—not 14 calendar days

GROUTED NODES NEED CURING

Full dry assembly and lossless reversal are not established.

Sources: project build-up and cost workbook (P3, P4). Labour, curing, transport and lifting assumptions require confirmation.

Useful shade—with a clear path to validation

VineWeave Canopy integrates form, planting and a proposed build-up.

01 AN INTEGRATED SUPPORT

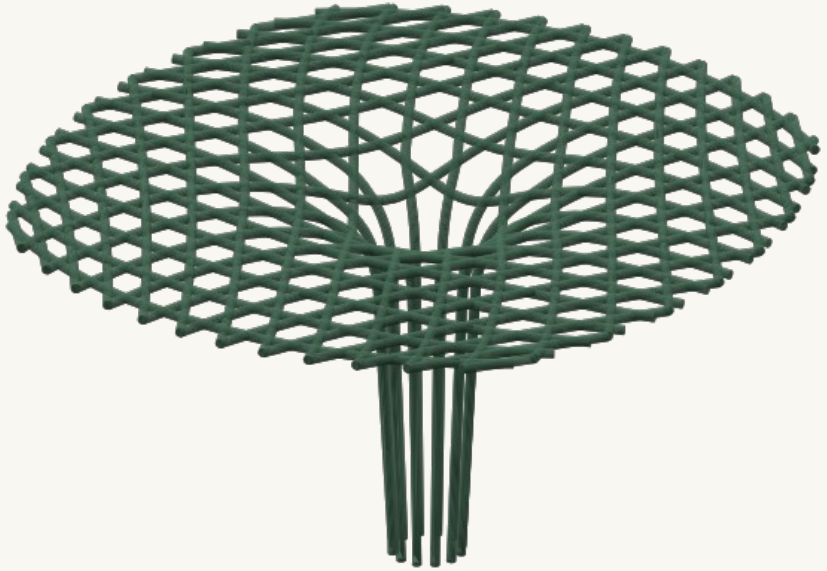
Tree-like geometry and climbing vegetation for local shade.

02 EXPLICIT PERFORMANCE SCENARIOS

Geometry, cover and sun position jointly shape useful blockage.

03 A DEFINED NEXT STEP

Confirm products, joints and site conditions; then validate outcomes.



Project synthesis (P1–P5). Scenario evidence supports the concept, not an engineering or measured-comfort certification.

Sources and evidence boundaries

Published-source leads and supplied project records are distinguished.

[1] OUTDOOR SHADE & COMFORT

Watanabe et al. (2014). Evaluation of outdoor thermal comfort in sunlight, building shade, and pergola shade during summer in a humid subtropical region. Building and Environment, 82, 556–565.

[2] GREENING MECHANISMS

Hoelscher et al. (2016). Quantifying cooling effects of facade greening: Shading, transpiration and insulation. Energy and Buildings, 114, 283–290.

[3] SOLAR GEOMETRY

NOAA Global Monitoring Division. General Solar Position Calculations.
gml.noaa.gov/grad/solcalc/solareqns.PDF

[4] COMFORT METHOD

UTCI Calculator / COST Action 730.
www.utci.org/utci_calc.php
Method reference only; no project UTCI result is claimed.

PROJECT RECORDS

P1 Model & take-off • P2 Ideal shade analysis • P3 Cost-Analysis.xlsx
P4 Proposed design & build-up • P5 Parameter and spatial-cover simulation

Unverified product factors, growth duration, actual joint restraint and field thermal response remain open.

Literature details were supplied with the project; original-source verification remains pending.